

Toward a Microstructural Derivation of Cascade Parameters in Rough/Multifractal Volatility

finprobs Research Project

July 2026

Abstract

El Euch et al. [2018] showed that rough volatility ($H > 0$) emerges as a scaling limit of order flow modeled by a nearly-unstable Hawkes process. No analogous derivation exists for the multifractal/cascade endpoint ($H \rightarrow 0$) of the log S-fBM model, or for its intermittency parameter λ^2 . We test and rule out three specific candidate mechanisms – linear Hawkes superposition, quadratic (Zumbach-effect) Hawkes feedback, and exact (rather than near-) criticality – each with an explicit proof or citation to rigorous prior theory. We survey non-Hawkes alternatives and find none connects rigorously to order-flow primitives either. We narrow the open problem to one precisely specified missing object: a joint double-scaling limit theorem, and identify a suggestive but unproven connection to existing results on the multiplicative chaos of $H = 0$ fractional Brownian fields.

1 Background and motivation

The log S-fBM covariance [Wu et al., 2022] is

$$C_\omega(\tau) = \frac{\nu^2}{2} [T^{2H} - \tau^{2H}] \xrightarrow[H \rightarrow 0, \lambda^2 \text{ fixed}]{} \lambda^2 \log(T/\tau). \quad (1)$$

The right-hand limit is a log-correlated Gaussian field, the object underlying Gaussian multiplicative chaos (GMC) [Kahane, 1985, Rhodes and Vargas, 2014] and the classical Multifractal Random Walk (MRW) of Bacry and Muzy [2003]. On the microstructural side, Jaisson and Rosenbaum [2016] and El Euch et al. [2018] show that a single nearly-unstable Hawkes process with heavy-tailed kernel $\varphi(x) \sim x^{-(1+\alpha)}$, $\alpha \in (1/2, 1)$, converges under joint rescaling to a rough Volterra/fractional CIR process with $H = \alpha - \frac{1}{2} \in (0, \frac{1}{2})$. This is a genuine derivation of H from order-flow primitives – not a fitted parameter.

Finding 1.1. No paper derives the cascade/multifractal parameters (λ^2, T) of the log S-fBM family from any microstructural mechanism. This is an established gap in the literature, not merely an unverified claim – we searched specifically and found nothing.

We test three natural candidate mechanisms for closing this gap and survey a fourth, non-Hawkes, category.

2 Candidate mechanism 1: linear Hawkes superposition

Proposition 2.1. *Let $N_t = \sum_k N_t^{(k)}$ be a superposition of independent, or jointly Gaussian-limiting, nearly-unstable Hawkes processes at distinct timescales, each converging via Jaisson and*

Rosenbaum [2016] to a Gaussian process $X_t^{(k)}$ with Hurst exponent H_k . If $\log \sigma_t = \sum_k a_k X_t^{(k)}$ for fixed (time- and state-independent) weights a_k , then $\log \sigma_t$ is Gaussian, and its structure function is exactly linear: $\zeta(q) = qH^*$ for some effective H^* .

Proof. A linear combination of jointly Gaussian random variables is Gaussian. For a self-similar Gaussian process with stationary increments, $\mathbb{E}|\Delta_\tau \log \sigma|^q = c_q \text{Var}(\Delta_\tau \log \sigma)^{q/2}$ by the standard normal absolute-moment identity, and $\text{Var}(\Delta_\tau \log \sigma) \sim \tau^{2H^*}$ by self-similarity of the dominant component at long lag. Hence $\zeta(q) = qH^*$, linear in q . $\square$

Finding 2.2. Negative result (with two caveats). Linear superposition can produce rich long-range dependence (an effective H^* shaped by the mixture of timescales) but cannot produce genuine multifractal curvature in $\zeta(q)$. **Caveat A:** this requires *jointly* Gaussian components with time-invariant weights; if a_k depend on regime or on another band’s realized level, the sum is generically non-Gaussian and the conclusion fails. **Caveat B:** the negative result concerns $\log \sigma_t$ itself, not σ_t or returns. If $\log \sigma_t$ is Gaussian with a *log-correlated* covariance – exactly (1) – then $\sigma_t = e^{\log \sigma_t}$ is already genuinely multifractal via the standard GMC mechanism, without requiring any hierarchy. This reframes the problem: the question is not “how to escape Gaussianity” but “how to make a single microstructural mechanism produce log-correlated, rather than power-law, Gaussian log-volatility.”

3 Candidate mechanism 2: quadratic (Zumbach) Hawkes

Quadratic Hawkes processes [Blanc et al., 2017] let the intensity depend on a squared, weighted average of past price moves, $\lambda_t = \lambda_\infty + H_t + Z_t^2$, capturing the empirically documented Zumbach effect (trends raise future volatility more than mean-reversion does). Dandapani et al. [2021] extend this with long-memory kernels to a “super-Heston” rough volatility limit with enhanced vol-of-vol.

Finding 3.1. Negative result. The quadratic feedback genuinely escapes Gaussianity, but into the *affine/Pearson-diffusion* (CIR/Heston-type) universality class, not the log-normal cascade class. We checked Dandapani et al. [2021] directly for multifractal-spectrum or $\zeta(q)$ -curvature content and found none; the underlying object is structurally a square-root/Feller-type diffusion, which does not exhibit the self-similar log-covariance decay a genuine cascade requires. This is a useful and rigorous microstructural derivation of the Zumbach/leverage effect, but a dead end for the specific question of cascade-parameter derivation.

4 Candidate mechanism 3: exact criticality

We next ask whether *exact* (rather than merely near-) criticality of the Hawkes branching ratio changes the qualitative scaling regime.

Proposition 4.1 (Cohen and Dombry, 2024). *For a Hawkes process with branching ratio fixed at exactly 1: if offspring dispersion is finite-variance, the rescaled intensity converges to a CIR process without mean-reversion, $d\lambda_t = \kappa dt + \sigma\sqrt{\lambda_t} dW_t$, for which $\text{Var}(\lambda_t) \sim \kappa\sigma^2 t^2/2$ – explosive, super-linear growth. If offspring dispersion is heavy-tailed, the process instead behaves subcritically, with power-law long-range dependence.*

Classical Yaglom-type theory [Yaglom, 1947] gives a third regime: a critical branching process conditioned on survival has population size scaling *linearly* in t .

Finding 4.2. Negative result. None of the three known regimes under exact criticality – explosive $\sim t^2$ growth, power-law long-range dependence, or linear Yaglom-type growth – is logarithmic. We flag an important conceptual distinction: “criticality” of a Hawkes branching ratio in *time* and “criticality” of a multiplicative cascade across *dyadic scales* (the actual mechanism behind (1) and GMC) are different notions, and should not be conflated. The number of independent-variance cascade steps needed to resolve scale τ in a dyadic construction is $\sim \log_2(T/\tau)$ – this is where the logarithm in (1) actually comes from, and it has no counterpart in Hawkes branching-ratio criticality.

5 Non-Hawkes alternatives

We surveyed three further categories.

- **Kesten-type multiplicative recursions** [Sornette and Cont, 1997] are a classical route to power-law tails, but we found no paper casting order-book depth or liquidity as a Kesten variable feeding a genuine nested-timescale cascade; existing stochastic-liquidity models treat depth as an exogenous diffusion, not a self-referential multiplicative recursion.
- **Branching random walks and GMC constructions** [Rhodes and Vargas, 2014] rigorously construct log-correlated fields from branching processes, and Simatos [2014] rigorously couples a branching random walk to a limit-order-book model – but for a different (price-extremes) question, with no connection to a log-correlated volatility field.
- **Agent-based/cascade trader models** [Ghashghaie et al., 1996] propose a turbulence-cascade analogy, disputed on methodological grounds by Barndorff-Nielsen and Prause [1997], and later hierarchical fits [Various authors, 2018] remain top-down statistical fits with ad hoc corrections, not order-flow derivations.

Finding 5.1. Negative result. No non-Hawkes mechanism found in the literature rigorously connects to order-flow primitives either. Everything outside Hawkes is either pure mathematics with no market microfoundation, or phenomenological analogy fitted top-down to data.

6 A precisely specified open problem

Hager and Neuman [2020] show that properly renormalized fractional Brownian motion itself converges, as $H \rightarrow 0$, to exactly the log-correlated Gaussian field / GMC structure needed for (1) – via pure Gaussian-field renormalization, with no microstructure underneath. This is suggestive: the target object is already known to arise as an $H \rightarrow 0$ limit of the *same family* that Hawkes processes already converge to at $H > 0$.

Open Problem 6.1. Does the established single-Hawkes scaling limit of Jaisson and Rosenbaum [2016] (kernel tail index $\alpha \rightarrow \frac{1}{2}$ from above, i.e. $H \rightarrow 0$) connect directly to the Hager and Neuman [2020] log-correlated limit, without requiring a multiplicative hierarchy? We searched the literature for treatment of the $\alpha \leq \frac{1}{2}$ boundary and found none – existing proofs rely on square-integrability of the limiting Riemann-Liouville kernel $g(t) = t^{\alpha-1}/\Gamma(\alpha)$, which is lost exactly at $H = 0$. This is a genuine technical wall, not evidence either way.

We state precisely what would be needed: a functional limit theorem for a nearly-unstable Hawkes process with kernel tail index $\alpha_n \rightarrow \frac{1}{2}$ *jointly* with the subcriticality gap $(1 - \|\varphi_n\|_1) \rightarrow 0$,

at a coupled rate, under which the rescaled *log*-intensity – not the intensity itself – converges to a Gaussian process whose covariance degenerates from power-law to logarithmic. No such theorem currently exists in the finance or point-process literature we searched.

7 Findings summary

- **Established.** Rough volatility ($H > 0$) is derived from Hawkes microstructure [El Euch et al., 2018, Jaisson and Rosenbaum, 2016]; no analogous derivation exists for cascade parameters.
- **Negative result.** Linear Hawkes superposition (Proposition 1), with two precisely stated caveats.
- **Negative result.** Quadratic/Zumbach Hawkes (wrong universality class).
- **Negative result.** Exact criticality (three known regimes, none logarithmic).
- **Negative result.** All non-Hawkes mechanisms surveyed.
- **Speculative.** The $\alpha \rightarrow \frac{1}{2}$ boundary connection to Hager and Neuman [2020], with the missing theorem precisely specified (Open Problem 1).

8 Future directions

1. Attempt the joint double-scaling limit theorem specified in Open Problem 1, drawing on the Hager and Neuman [2020] renormalization as a template for the target object.
2. Investigate whether a Cox-Hawkes hierarchy with genuinely multiplicative (exponential) cross-scale modulation – as opposed to the ruled-out linear superposition and quadratic feedback – can supply the missing log-covariance structure; the specific missing lemma is a scaling-limit theorem controlling the exact log-covariance decay across levels, not merely aggregate intensity fluctuations.
3. Even if a positive theorem is found, assess estimability: participant-segmented tick data is not publicly available, so any resulting mechanism may remain empirically unfalsifiable at the microstructure level with realistically obtainable data.

Acknowledgments

This paper synthesizes research conducted by an orchestrated multi-agent AI investigation, with all citations independently re-verified against primary sources.

References

- E. Bacry and J.-F. Muzy. Log-infinitely divisible multifractal processes. *Communications in Mathematical Physics*, 236(3):449–475, 2003.
- O. E. Barndorff-Nielsen and K. Prause. Comment on “Turbulent cascades in foreign exchange markets”. *cond-mat/9607120*, 1997.

- P. Blanc, J. Donier, and J.-P. Bouchaud. Quadratic Hawkes processes for financial prices. *Quantitative Finance*, 17(2):171–188, 2017.
- S. N. Cohen and C. Dombry. Functional limit theorems for Hawkes processes. *Probability Theory and Related Fields*, 2024. arXiv:2401.11495.
- A. Dandapani, P. Jusselin, and M. Rosenbaum. From quadratic Hawkes processes to super-Heston rough volatility models with Zumbach effect. *Quantitative Finance*, 21(8):1235–1247, 2021.
- O. El Euch, M. Fukasawa, and M. Rosenbaum. The microstructural foundations of leverage effect and rough volatility. *Finance and Stochastics*, 22(2):241–280, 2018.
- S. Ghoshghaie, W. Breymann, J. Peinke, P. Talkner, and Y. Dodge. Turbulent cascades in foreign exchange markets. *Nature*, 381:767–770, 1996.
- K. Hager and E. Neuman. The multiplicative chaos of $H = 0$ fractional Brownian fields. *arXiv:2008.01385*, 2020.
- T. Jaisson and M. Rosenbaum. Rough fractional diffusions as scaling limits of nearly unstable heavy tailed Hawkes processes. *Annals of Applied Probability*, 26(5):2860–2882, 2016.
- R. Rhodes and V. Vargas. Gaussian multiplicative chaos and applications: a review. *Probability Surveys*, 11:315–392, 2014.
- F. Simatos. Coupling limit order books and branching random walks. *Journal of Applied Probability*, 51(3):625–639, 2014.
- D. Sornette and R. Cont. Convergent multiplicative processes repelled from zero: power laws and truncated power laws. *cond-mat/9609074*, 1997.
- Multiplicative random cascades with additional stochastic process in financial markets. *arXiv:1809.00820*, 2018.
- Q. Wu, J.-F. Muzy, and E. Bacry. From rough to multifractal volatility: the log S-fBM model. *Physica A*, 604:127919, 2022. arXiv:2201.09516.
- A. M. Yaglom. Certain limit theorems of the theory of branching random processes. *Doklady Akademii Nauk SSSR*, 56:795–798, 1947.
- J.-P. Kahane. Sur le chaos multiplicatif. *Annales de la Faculté des Sciences de Toulouse*, 9(2):105–150, 1985.